Kinetix

The AI-driven post-op ecosystem that guides, tracks, and gamifies your home rehabilitation.

Transforming physical therapy with **Computer Vision** and **NLP** — built for the future of remote recovery.

Why Home Rehab Fails

😴 Zero Motivation

Patients lose interest and abandon therapy routines without feedback loops or reward systems.

⚠️ Incorrect Form

Unsupervised exercise leads to dangerous compensatory movements and re-injury risk.

📉 Zero Doctor Visibility

Clinicians have no real-time data on daily compliance, pain levels, or recovery trajectory.

🖐️ Physical Barriers

Post-op patients struggle to tap, swipe, or type — standard apps are simply inaccessible.

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THE SOLUTION

Meet Kinetix

A comprehensive, full-stack healthcare ecosystem that **brings the clinic into the patient's living room** — no wearables, no hardware, no barriers.

- Zero-touch voice navigation
- Real-time AI form correction
- Automated clinical summaries

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Feature 1 — RehabVision & Gamification

Live Pose Tracking

In-browser **Google MediaPipe Pose** renders a real-time skeletal overlay and calculates critical joint angles like knee flexion — directly in the browser, no backend latency.

Auto-Rep Counting

The system intelligently validates form before counting each rep — ensuring only **correct repetitions** are logged to the patient record.

Gamified Recovery

A **"Daily Recovery Velocity" score** and visual celebrations reward consistent effort, dramatically reducing patient drop-off rates.

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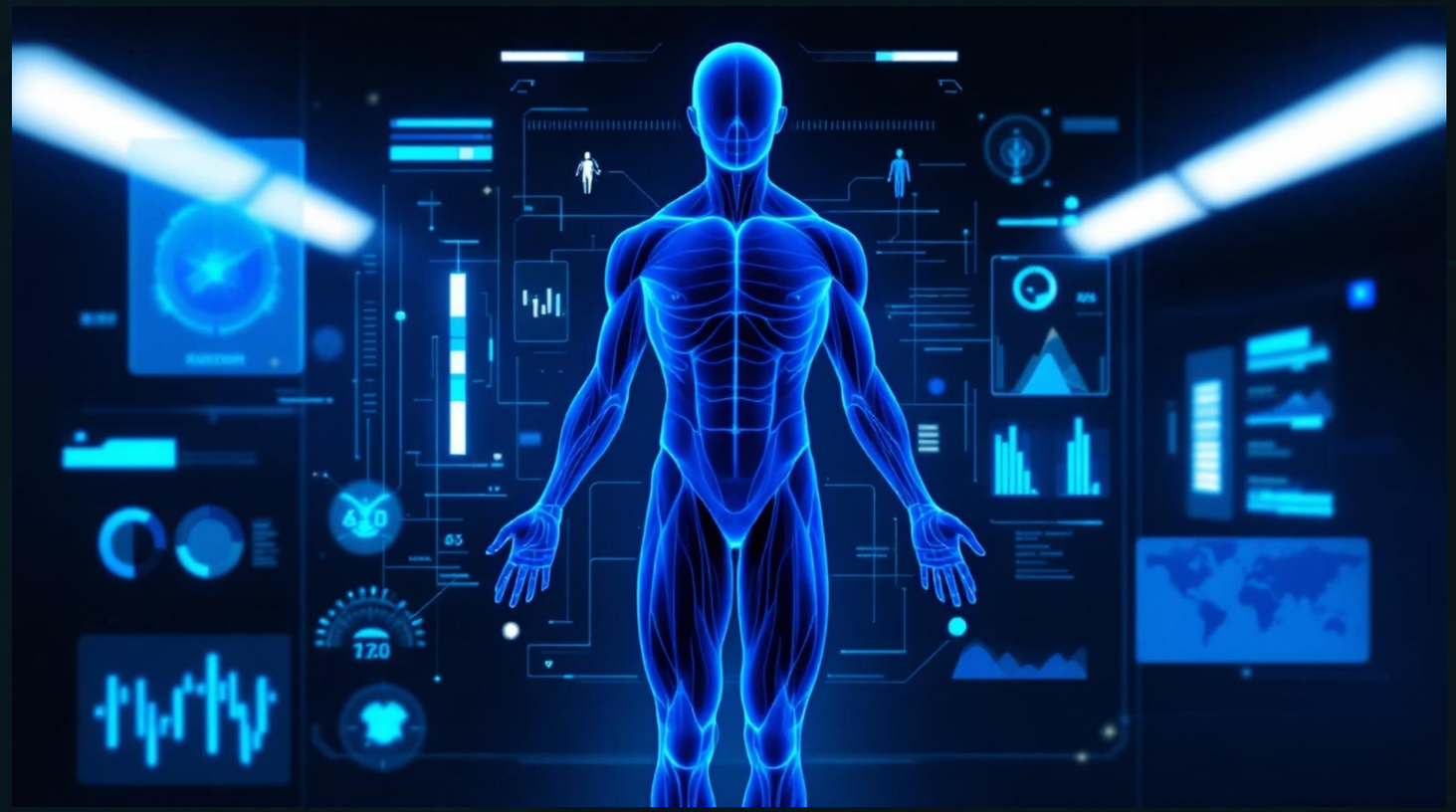

FEATURE 2

Interactive Pain Mapping

Goodbye to frustrating dropdown menus. Kinetix features a **polished, clickable visual map of the human muscular system** — patients simply tap the exact muscle or joint where they feel discomfort.

Anatomically Precise

Pinpoint specific muscles, not vague body regions.

Pain Intensity Logging

Records severity directly to the Supabase database in real time.

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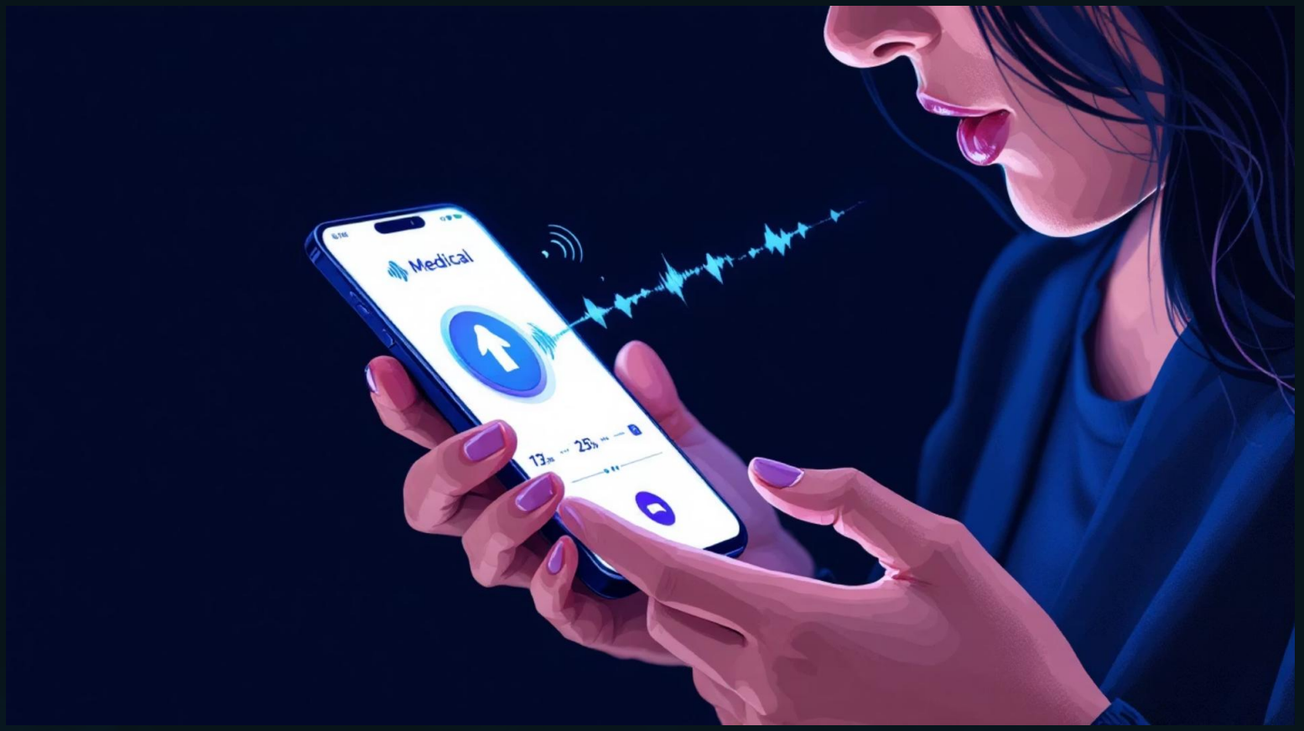
FEATURE 3

Hands-Free Voice Navigation

Built from the ground up for post-op accessibility. Patients with limited dexterity or bandaged hands simply speak to navigate the entire app — no touch required.

> *"Start my exercises"* — one spoken command triggers full UI state changes and launches the rehab session.

Powered by the native **Web Speech API** — zero third-party dependencies, instant response, works on any modern browser.



Feature 4 — AI Clinical Summaries for Doctors

📊 Aggregated Weekly Data

Pulls reps completed, missed sessions, pain trends, and recovery velocity scores automatically — no manual chart review.

🤖 LLM-Powered Synthesis

OpenAI / Gemini generates a concise, professional **2-sentence clinical summary** ready for the doctor's dashboard in milliseconds.

⏱️ Time Saved

Physicians spend less time deciphering raw data and more time making **high-value clinical decisions**.

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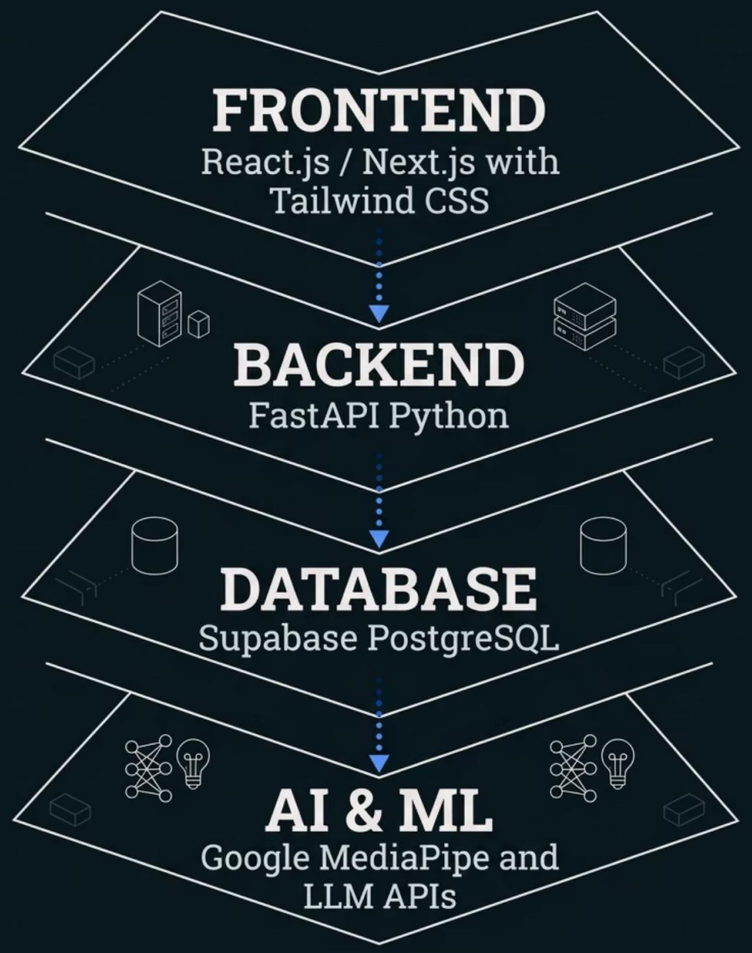
Under the Hood — Tech Stack



Purpose-Built for Performance

Every technology in Kinetix was chosen deliberately — **React/Next.js** for snappy client-side rendering, **FastAPI** for sub-millisecond Python routing, **Supabase** for HIPAA-aligned data security, and **MediaPipe** running entirely client-side to eliminate AI inference latency.

All Computer Vision runs **in-browser** — zero patient video data ever leaves the device.

The Golden Path — Live Demo

Voice Login
Patient says their name to authenticate

Pain Logging
Tap body map to record today's soreness

RehabVision
Perform live squats tracked and counted by AI

Doctor Dashboard
View instantly generated AI clinical summary

Four seamless interactions. One unified system. This is the **full patient-to-clinician loop** — demonstrated live in under 3 minutes.

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Kinetix — Bridging Hospital & Home

🌐 Scalable

Cloud-native architecture ready to serve thousands of concurrent patients across any device, anywhere.

♿ Accessible

Voice-first, touch-free design ensures no post-op patient is left behind regardless of physical limitation.

🚀 Ready to Ship

Full-stack, fully integrated, and demo-ready. Kinetix is built to revolutionise remote recovery — **today**.

Questions? We're ready. **Let's talk about the future of rehabilitation.**